

Jahangirnagar University



Assignment Topic

Case Study Analysis: Step Wise Planning for JU Comprehensive Student Information & Payment System.

Course Title: Software Project Management and Quality Assurance
Course Code: PMSCS 660

SUBMITTED TO

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PART-A

Conceptual Understanding

Question 1:

a) Aspirations of step wise:

i) practicality: The plan must be realistic and achievable within the current budget and team capacity.

ii) Scalability: The framework should work for both small and massive enterprise system.

iii) Range of Applications: It can be adapted for any type of software, from e-commerce to banking systems.

b) Stepwise vs PRINCE2:

i) Similarities: Both focus heavily on product-based planning. (like using product breakdown structures).

ii) Differences: PRINCE2 covers the entire project lifecycle including execution and monitoring.

Question 2:

c) Project authority vs. other stakeholders:

The project authority makes the final decisions and controls the budget. Other stakeholders are affected by the project but do not have decision making power.

d) revising objectives via stakeholder analysis:

Sometimes we need to change objectives to get key stakeholders on our side.

Example: If IT staff worried about the system crashing their servers, we might add a new objective: "Implement automated load balancing" to ensure their cooperation.

Question 3:

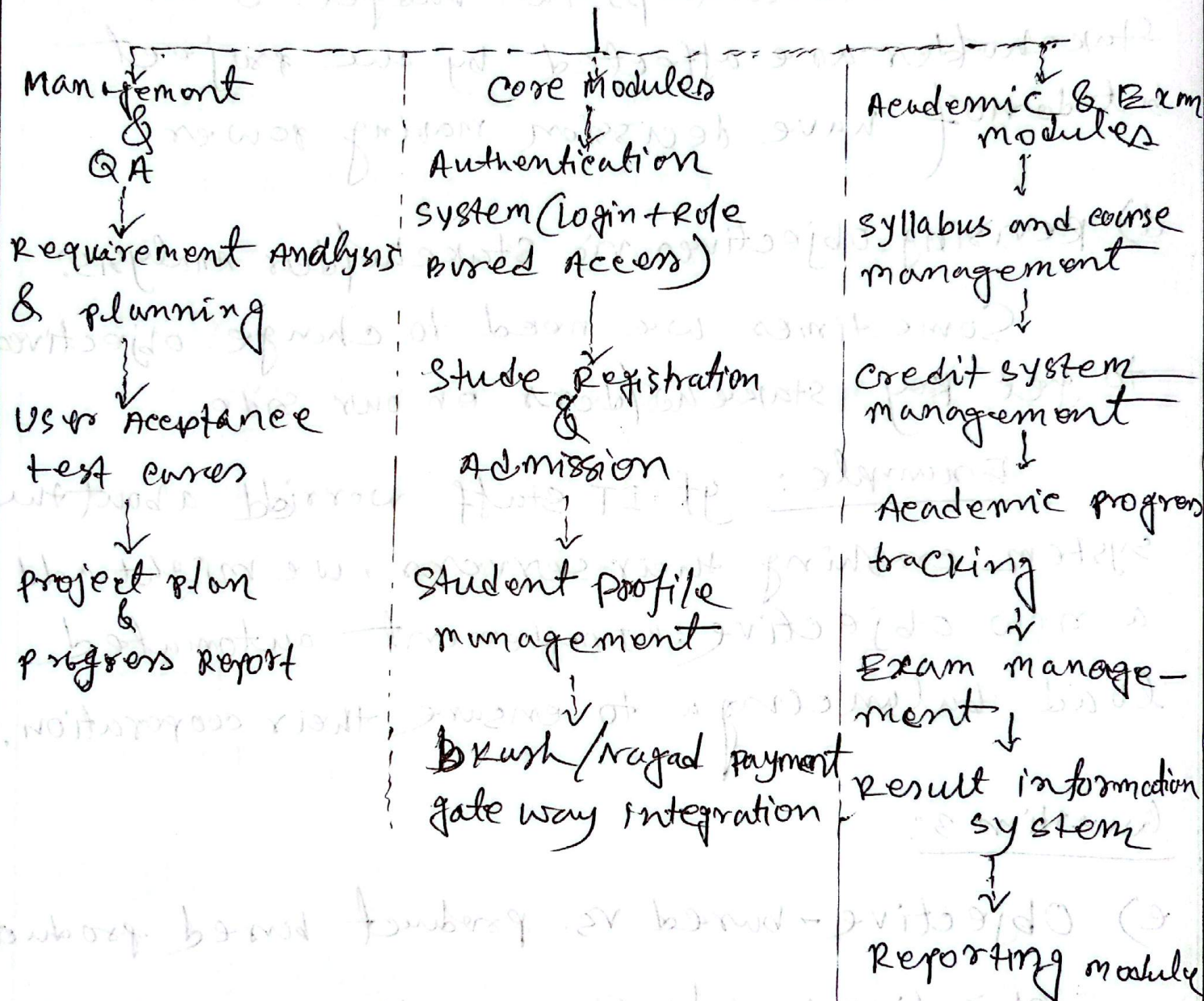
e) Objective-based vs. product-based product

Objective-based: We know the problem but not the exact software solution. Need agile.

Product-based: The final product is clearly defined from the start. A structure waterfall model works here.

f) product Breakdown structure (PBS) for JV Reg:

JV online student information system



Question 4:

Risk Reduction vs Contingency: Risk reduction means taking steps to prevent a problem. Contingency is the backup plan if the problem still happens.

4) Difference between Risk Reduction and contingency planning:

Risk reduction: preventing actions to decrease the probability of risk occurring or reduce its impact before it happens.

contingency planning: This means preparing alternative actions that will be taken if the risks actually happens.

Differences:

- 1) Risk reduction is proactive
- 2) Contingency planning is reactive.

4) Risk Reduction Activity and contingency Action for each risks:

Risks	Risk reduction Activity	contingency Activity
Team member refuse to complete psychometric questionnaires.	1) explain the purpose and benefits of the questionnaires.	Use alternative like, observation interviews and team assumptions.
Project managers are unwilling to use the applications.	Involve project manager in requirement gathering, demonstrations.	provide additional training and support and modify feedback

Example:

1. Questionnaires refusal: Explain benefits and ensure privacy for reduction method, and Contingency Conduct interview instead.
2. Managers unwilling to use applications
 - i) Reduction: involve manager development
 - ii) Contingency: provide training and modify features.

PART-B

Applied problem solving

Question 5:

Stepwise applied to JV system

Step 0: select project:

Action: prioritize this system over the university projects.

Artefacts: Approved project mandate.

challenge: secure the required budget and approval from JV syndicate.

Step 1: Identify project infrastructure.

Action: Set up technical environment and structure the 12 member team

Artefacts: Architecture doc~~ment~~ and team chart.

challenge: complying with BRTE BTRC rules to host student and financial data on local BP server

Step 3: Analyze project products & Activities

Action: Break down deliverables and

Artefacts: selected life cycle model.

challenge: Ensuring strict database accuracy for bKash and Agrani Bank financial settlements.

Step 4: Identify Project product & Activities:

Action: Break down deliverables and sequence the development tasks.

Artefacts: product breakdown structure and Activity network.

challenge: converting complex JV academic ordinances.

Step 5: Estimate Effort for Each Activity

Action: calculate the time and staff effort needed for each task.

Artefacts: Effort Estimation sheet.

challenge: Heavy traffic to spikes during JV course registration days.

Step 6: Identify Activity Risks:

Action: quantity risks and prepare contingency plans.

Artefacts: RISKS Register and contingency plan.

Challenge: Fintech API timeout.

Step 7: Allocate resources

Action: Assign the 12 cross functional team members to specific tasks.

Artefacts: project Gantt chart with resource names.

Challenges: Sequencing tasks properly so that frontend developers do not sit idle waiting for the complex backend payment APIs to be diminished.

Question 6:

1) Elicit distinct activities with justifications.

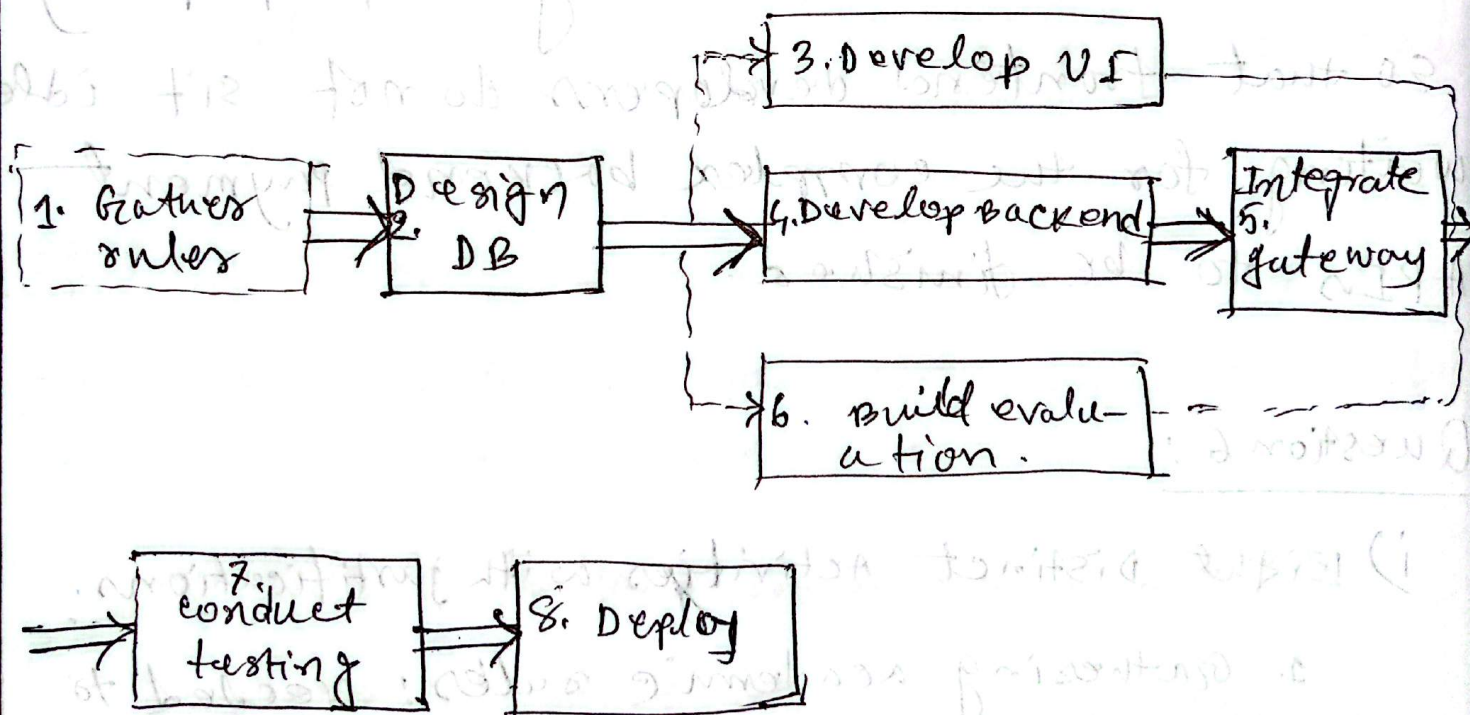
1. Gathering academic rules: Needed to understand JGU's specific regulations.

2. Design data base schema: security structure student, course and teachers.

3. Develop frontend UI: Required to create web interface for student and teachers.

4. Develop backend API: Acts as the core engine to process business logic and data
5. Integrate mobile gateway.
6. Build evaluation logic.
7. Conduct system testing.
8. Deploy to live servers.

j) Activity network (precedence diagram)
& critical path:



Legend:

→: Normal path; ⇒: critical path

critical path sequence: 1 → 2 → 4 → 5 → 7 → 8

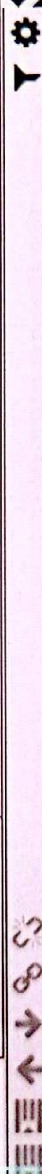
Total Estimated project duration: 12 weeks.

K) Gantt Project Data Table

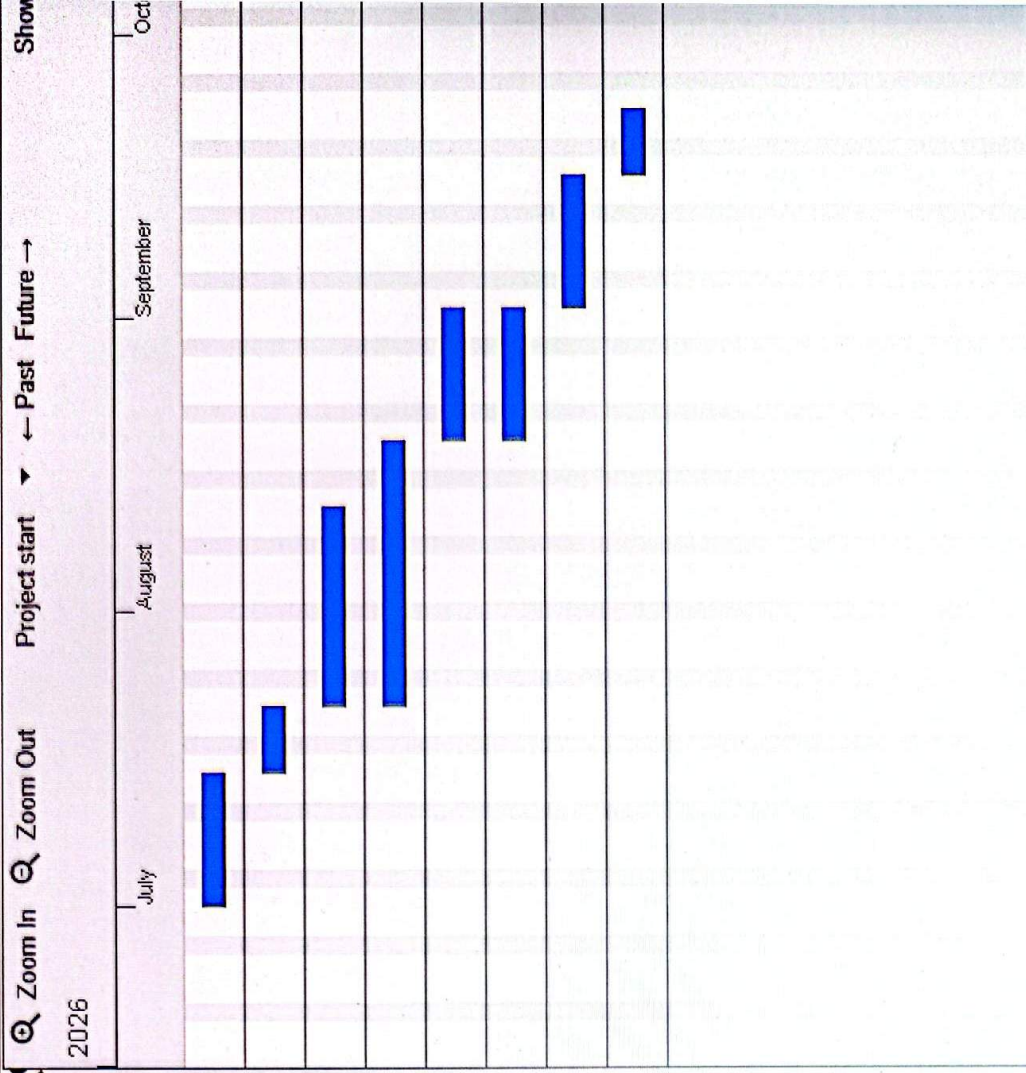


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Resources Chart



Name	Main Stakeholder	Begin date	End date
Gather JU academic rules	Rifat (Business Analyst)	7/1/26	7/14/26
Design database schema	Imtiaj & Rifat	7/15/26	7/21/26
Develop frontend UI	Rifat (UI Developer)	7/22/26	8/11/26
Develop backend APIs	Imtiaj (Backend Dev)	7/22/26	8/18/26
Build class evaluation logic	Imtiaj (Backend Dev)	8/19/26	9/1/26
Integrate bKash/Nagad gateway	Imtiaj (Backend Dev)	8/19/26	9/1/26
Conduct system testing	QA Team	9/2/26	9/15/26
Deploy to JU live servers	Imtiaj & Rifat	9/16/26	9/22/26



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PART-C

critical Analysis

Question 2:

l) Evaluation prototype vs waterfall:

waterfall is rigid and only works if requirement never change. Evaluation prototype is better when users are unsure of what they want as it follows them to test early versions and give feedback to improve the system gradually.

m) prototyping in the class scenario:

RISK 1 (confusing a 2): A prototype lets users test the system early so developer can fix the design before final coding.

RISK 2 (wrong report): HR can view sample output reports and request format change early.

n) Life cycle and Resource Allocation:

In waterfall, resources are used in stages (developers sit idle during planning, testers wait during coding). In an iterative model, resources must be allocated continuously, requiring more dynamic team schedule.

Question 8:

O) scalability in BD Government projects:
stepwise is scalable but needs adjustment for large BD government projects, we would need to expand to step 2 to strictly follow the Bangladesh National Digital Architecture (BNDA) guidelines. The project must also be broken into smaller "branches" to deliver public services in phases rather than waiting years for completion.

P) Step wise compatibility with Agile:
Stepwise is a traditional planning method and struggles with agile because it demands too much upfront planning. To use it in an agile environment,

Retain: Step 0-3 to establish the project vision.

Modify: Step 4-5 should only be applied to short 2-week sprints, not the whole project

Discard: Step 4.4 (long-term activity networks), since Agile paths iteratively, not all at once.